

Aim

To implement a data cube for a data warehouse on 3-dimensional data for product sales over time across different regions (product, date, country) using python

Algorithm:

1. Import libraries & define data structures (Use dataclass to model each product record.)
2. Define the Product entity (contains name (str), category(str) and price(float))
3. Instantiate dimension members (
 - Create a list of Product objects (name, category, price).
 - Create lists for all Date values and Country values)
4. Generate the fact table
5. Load facts into a pandas DataFrame
6. Create the 3-D data cube (pivot table)
7. Explore / analyze (slice & dice , Rollup, Extract)
8. Visualize or integrate downstream - Plot with matplotlib or feed to a BI tool.
 - Store in a data-warehouse table or OLAP engine for large-scale analytics.

Code:

```
import pandas as pd
from dataclasses import dataclass

# Define the Product class
@dataclass
class Product:
    name: str
    category: str
    price: float

# Sample product data
products = [
    Product("Laptop", "Electronics", 1000),
    Product("T-shirt", "Clothing", 20),
    Product("Book", "Books", 15),
    Product("Headphones", "Electronics", 100),
    Product("Jeans", "Clothing", 50),
    Product("Smartphone", "Electronics", 800),
    Product("Sunglasses", "Accessories", 30),
    Product("Watch", "Accessories", 50),
    Product("Shoes", "Footwear", 80),
]
```

```

# Define dates and countries
dates = ["2023-05-01", "2023-05-02", "2023-05-03"]
countries = ["USA", "UK", "Germany"]

# Generate the full dataset
data = []
for product in products:
    for date in dates:
        for country in countries:
            data.append({
                "Category": product.category,
                "Product": product.name,
                "Date": date,
                "Country": country,
                "Price": product.price
            })

# Convert to DataFrame
df = pd.DataFrame(data)

# Create a pivot table (data cube)
data_cube = pd.pivot_table(
    df,
    values="Price",
    index=["Category", "Product"],
    columns=["Date", "Country"],
    aggfunc="first" # each price is unique and atomic
)

# Display the data cube
print("3D Data Cube (Product x Date x Country):\n")
print(data_cube)

```

```

# sliceup operation
# Selecting a single value for one dimension, reducing the cube's
dimensionality.
# how Laptop prices across countries and dates
print(data_cube.loc[("Electronics", "Laptop")])

# Rollup operation
# Definition: Aggregating data up a hierarchy or grouping dimension.
# Roll-up by Product Category → Sum of prices for each category.

```

```
print(data_cube.groupby(level=0).sum())

#Export
data_cube.to_csv("MyDrive\cube.csv")
```

Result: cube is a compact 3-dimensional structure that supports fast analytical queries (slice, dice, roll-up, drill-down) using familiar pandas operations.